COMPSCI 713: AI Fundamentals

Lecture 4: Logic Neural Networks

Bridging Symbolic Logic and Neural Networks for AI Reasoning

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What You Will Learn Today?

- Why Logic in AI?
- Recap on symbolic logic
- How Do Neural Networks Work?
- The Need for Logic Neural Networks
- Differentiable Logic – Making Logic Work in AI
- Understanding Logic Neural Networks (LNNs)

Why Logic in AI?

AI Needs More Than Just Neural Networks

Traditional AI Challenges:

- Deep learning models excel at pattern recognition but lack reasoning ability.
- AI cannot explain its decisions (**black-box problem**).

Logic provides:

- **Explicit reasoning:** If $A \to B$ and A is true, then B must be true.
- **Explainability:** We can see *why* a decision was made.

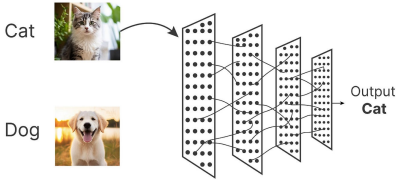
Why Logic in AI?

Example:

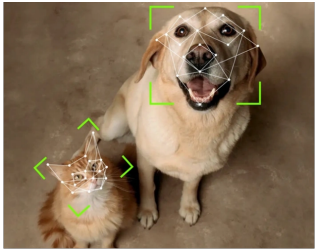
A neural network can recognize a cat, but can it explain *why*?

Logic-based AI can provide structured reasoning:

"If it has a smaller size, with fur → it is likely a cat."



Classification Through
Convolutional Neural Network



Classification Through
Logical Reasoning

Recap on Symbolic Logic – Propositional Logic

Key Components of Propositional Logic

Deals with statements that are either **True** or **False**.

- Uses logical operators to form complex expressions.

Logical Operators and Their Meanings:

Operator	Symbol	Meaning
Negation (NOT)	$\neg P$	Negation (opposite)
Conjunction (AND)	$P \wedge Q$	True if both P and Q are true
Disjunction (OR)	$P \vee Q$	True if at least one is true
Implication	$P \rightarrow Q$	False if P is true, Q is false; others, true
Equivalence	$P \leftrightarrow Q$	True if P and Q are either both true or both false

Truth Table for Conjunction ($\wedge$)

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: "It is raining AND I have an umbrella" $\rightarrow$ This is only True if *both* statements are True.

Exercise 1: Which of the following statements is true?

A) If P is false, then $P \wedge Q$ is always true.
B) If P is true and Q is true, then $P \wedge Q$ is true.
C) If P is false and Q is true, then $P \wedge Q$ is true.
D) If P is true and Q is false, then $P \wedge Q$ is true.

Recap on Symbolic Logic – Propositional Logic

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D) If P is true and Q is false, then $P \wedge Q$ is true.

(Answer: B)

Recap on Symbolic Logic – Propositional Logic

Truth Table for Implies ($\rightarrow$)

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Why is $(F \rightarrow T)$ and $(F \rightarrow F)$ both True?

- An implication only fails when the first part (P) is true and the second part (Q) is false.
- When P is false, the implication is always true because we cannot verify whether Q should happen or not.
- This is called **"vacuous truth"**: a false premise makes the entire statement automatically true.

Understanding Implication in Real Life

Statement: "If it rains, the ground will be wet."

P: It rains. Q: The ground is wet. $P \to Q$: Whenever it rains, the ground must be wet.

Situation	P (Rains)	Q (Ground Wet)	$P \to Q$ (Valid?)
It rains and the ground is wet.	T	T	T
It rains but the ground is NOT wet.	T	F	F
It does NOT rain, but ground is wet (e.g. hose).	F	T	T
It does NOT rain and the ground is dry.	F	F	T

Exercise 2: Truth Table Reasoning

Given: The rule "If an animal is a bird, then it can fly" ($\text{Bird}(x) \rightarrow \text{CanFly}(x)$).

Which of the following situations **violates** this rule?

A) An eagle is a bird and it can fly.

B) A penguin is a bird, but it cannot fly.

C) A fish is not a bird, and it cannot fly.

D) A bat is not a bird, but it can fly.

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$Bird(x)$	$CanFly(x)$	$P \rightarrow Q$
T	T	T
T	F	F (violation)
F	T	T
F	F	T

(Answer: B, because the rule states ALL birds must fly, but penguins don't.)

Recap on Symbolic Logic – First-Order Logic

Key Components of First-Order Logic (FOL)

- Variables, relations, functions, quantifiers, signature, terms, sentences.
- Allows more complex relationships to be expressed.

Concept	Symbol	Meaning
Relation	$R(x, y)$	Describes a relationship (e.g. $\text{Parent}(\text{Alice}, \text{Bob})$)
Function	$f(x)$	Maps an entity to another (e.g. $\text{Age}(\text{Bob}) = 30$)
Universal Quantifier	$\forall x\ R(x, y)$	"For all x, $R(x, y)$ holds"
Existential Quantifier	$\exists x\ R(x, y)$	"There exists at least one x such that $R(x, y)$ holds"

Recap on Symbolic Logic – First-Order Logic

Exercise 3
Given Rule:

$$\forall x\ (\text{Student}(x) \rightarrow \text{Enrolled}(x, \text{University}))$$

"For all x, if x is a student, then x is enrolled in a university."

Question: Which of the following conclusions is **incorrect**?

A. $\text{Student}(\text{Bob}) \rightarrow \text{Enrolled}(\text{Bob}, \text{University})$

B. $\exists x\ (\text{Student}(x) \wedge \text{Enrolled}(x, \text{University}))$

C. $\neg\text{Enrolled}(\text{Bob}, \text{University}) \rightarrow \neg\text{Student}(\text{Bob})$

D. $\forall x\ (\text{Enrolled}(x, \text{University}) \rightarrow \text{Student}(x))$

Recap on Symbolic Logic – First-Order Logic

Exercise 3
Given Rule:

$$\forall x\ (\text{Student}(x) \rightarrow \text{Enrolled}(x, \text{University}))$$

"For all x, if x is a student, then x is enrolled in a university."

Question: Which of the following conclusions is **incorrect**?

A. $\text{Student}(\text{Bob}) \rightarrow \text{Enrolled}(\text{Bob}, \text{University})$
(If Bob is a student, then Bob is enrolled in a university.)

B. $\exists x\ (\text{Student}(x) \wedge \text{Enrolled}(x, \text{University}))$
(Some students are enrolled in a university.)

C. $\neg\text{Enrolled}(\text{Bob}, \text{University}) \rightarrow \neg\text{Student}(\text{Bob})$
(If Bob is not enrolled in a university, then Bob is not a student.)

D. $\forall x\ (\text{Enrolled}(x, \text{University}) \rightarrow \text{Student}(x))$
(All people enrolled in university are students.)

(Answer: D — the rule only states students → enrolled, not enrolled → students.)

Recap on Symbolic Logic – Logical Inference

Logical inference is the process of deriving conclusions from given statements (premises) using formal rules of logic.

There are several key rules of inference used in classical logic, but in this section, we will focus on three fundamental types:

- **Modus Ponens** – Direct inference using conditional statements.
- **Modus Tollens** – Reasoning using negation of the conclusion.
- **Syllogism** – Connecting multiple premises to draw a final conclusion.

Recap on Symbolic Logic – Logical Inference

Modus Ponens

- For any P, Q, $((P \to Q) \land P) \Rightarrow Q$.
- is a rule of inference that allows us to conclude the consequent Q from $P \to Q$, given that the antecedent P is true.

Pattern:

1. 1st Premise: $P \to Q$ (If P then Q)
2. 2nd Premise: P (P is true)
3. Conclusion: Q (Therefore, Q must be true)

Example: Weather and Wet Ground

- If it rains, the ground gets wet. (Rains $\to$ WetGround)
- It is raining. (Rains $=$ True)
- **Conclusion:** The ground is wet. (WetGround $=$ True)

Recap on Symbolic Logic – Logical Inference

Modus Tollens

- For any P, Q, $((P \to Q) \land \neg Q) \Rightarrow \neg P$
- is a rule of inference that allows us to conclude $\neg P$ if the consequent Q in $P \to Q$ is false.

Pattern:

1. 1st Premise: $P \to Q$ (If P then Q)
2. 2nd Premise: $\neg Q$ (Q is false)
3. Conclusion: $\neg P$ (Therefore, P must be false)

Example: Traffic Rules

- If the traffic light is green, cars can go. (GreenLight $\to$ CanGo)
- Cars cannot go. ($\neg$CanGo)
- **Conclusion:** The traffic light is not green. ($\neg$GreenLight)

Recap on Symbolic Logic – Logical Inference

Syllogism

- A logical argument that connects two premises to reach a conclusion.

Pattern:

1. 1st Premise: $A \to B$ (If A then B)
2. 2nd Premise: $B \to C$ (If B then C)
3. Conclusion: $A \to C$ (Therefore, if A then C)

Example: Animal Classification

- All dogs are mammals. $(\text{Dog}(x) \to \text{Mammal}(x))$
- All mammals have a backbone. $(\text{Mammal}(x) \to \text{HasBackbone}(x))$
- **Conclusion:** All dogs have a backbone. $(\text{Dog}(x) \to \text{HasBackbone}(x))$

Recap on Symbolic Logic – Logical Inference

Exercise 4: Evaluating Scientific Theories
Given Rules:

1. $\text{SupportedByData}(T) \rightarrow \text{AcceptedByCommunity}(T)$
2. $\text{AcceptedByCommunity}(T) \rightarrow \text{ReceivesFunding}(T)$
3. $\text{ReceivesFunding}(T) \rightarrow \text{RecognizedInAcademia}(T)$

Fact: $\neg\text{RecognizedInAcademia}(\text{QGTheory})$ (If The Quantum-Gravity Unification Theory is "not widely recognized in academia".)

Question: What can we conclude?

A) QG Theory is well-supported by experimental data.

B) QG Theory is not well-supported by experimental data.

C) QG Theory is accepted by the scientific community.

D) QG Theory receives funding for further research.

Recap on Symbolic Logic – Logical Inference

Exercise 4 – Answer: B

Applying **Modus Tollens** repeatedly:

1. **Step 1:** Rule: $\text{ReceivesFunding}(T) \rightarrow \text{RecognizedInAcademia}(T)$
   Fact: $\neg\text{RecognizedInAcademia}(\text{QGTheory})$
   Inference: $\neg\text{ReceivesFunding}(\text{QGTheory})$ (If a theory is not recognized, it does not receive funding.)
2. **Step 2:** Rule: $\text{AcceptedByCommunity}(T) \rightarrow \text{ReceivesFunding}(T)$
   Fact: $\neg\text{ReceivesFunding}(\text{QGTheory})$
   Inference: $\neg\text{AcceptedByCommunity}(\text{QGTheory})$ (If a theory does not receive funding, it is not accepted by the scientific community.)
3. **Step 3:** Rule: $\text{SupportedByData}(T) \rightarrow \text{AcceptedByCommunity}(T)$
   Fact: $\neg\text{AcceptedByCommunity}(\text{QGTheory})$
   Inference: $\neg\text{SupportedByData}(\text{QGTheory})$ (If a theory is not accepted, it is not well-supported by data.)

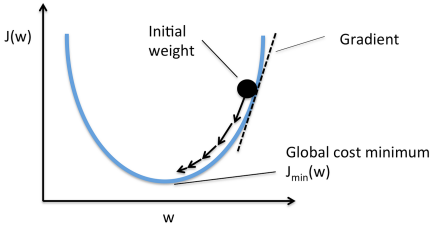
How Do Neural Networks Work?

How Does Training Work?

Training a neural network involves adjusting the weights to minimize the difference between predictions and correct answers.

Key Training Steps:

- **Forward Propagation:** Input passes through layers, generating an output.
- **Loss Calculation:** Difference between predicted and actual value (loss function).
- **Backpropagation:** Network adjusts weights by computing gradients to reduce errors.
- **Optimization using Gradient Descent:** Network updates weights step-by-step to improve accuracy.



References:

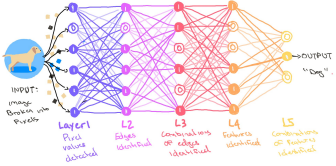
`http://neuralnetworksanddeeplearning.com/`
`https://d2l.ai/index.html`

How Do Neural Networks Work?

Example: Cat vs. Dog Classifier

Imagine we want a neural network to classify images as either cats or dogs

- **Step 1 – Input Layer:**
  Raw image is converted into pixel values (e.g. RGB values).
- **Step 2 – Hidden Layers:**
  The network extracts features (e.g., edges, fur patterns, eyes). Each layer learns more complex features (early layers detect edges, deeper layers detect shapes like ears).
- **Step 3 – Output Layer:**
  The network assigns a probability to each class: 80% Dog, 20% Cat → Predicted Label: Dog



BUT: Does the neural network understand what a cat is? **No!** It only detects patterns, unlike humans who reason symbolically.

The Need for Logic Neural Networks

Why Combine Logic with Neural Networks?

Limitations of Traditional Neural Networks:

- **Black-box problem:** Neural networks are powerful but lack explainability.
- **Data dependency:** Require huge amounts of labeled data.
- **Lack of logical reasoning:** Struggle with rule-based inference.
- **Error sensitivity:** Confident but incorrect predictions can occur.

Example:

- A neural network trained on medical images can predict diseases, but it **cannot explain why** a particular image suggests cancer.
- Doctors need an *explanation*, not just a prediction!

The Need for Logic Neural Networks

Why Logic Helps:

- **Add explainability:** Rules and constraints provide transparent reasoning.
- **Reduces data dependency:** Knowledge-based reasoning reduces the need for big data.
- **Improves consistency:** Logical constraints prevent contradictory outputs.

Example: An AI for loan approval must not only predict "Approved/Rejected" but also provide reasoning like: "Loan approved because salary is high and credit score is good."

Why Not Just Use Logic Alone?

- Logic systems are **hard to scale**: manually writing thousands of rules is impractical.
- Logic **struggles with uncertainty**: e.g. It cannot handle probabilistic situations like "How likely is it to rain tomorrow?"

The Solution

Logic Neural Networks (LNNs) combine the strengths of both!

Differentiable Logic – Making Logic Work in AI

Why Can't Neural Networks Use Traditional Logic?

- Logic is **discrete** (True/False), but neural networks work with **continuous values** (0-1).
- **Solution:** Convert logic into differentiable operations so AI can learn from it.

Example: Instead of strict Boolean logic (0 or 1), we use soft logic that assigns probabilities (e.g., 0.8 = likely true).

Soft Logic Operators (Product-Sum):

Logical Operation	Traditional (Boolean)	Soft Logic (Continuous)
AND ($\wedge$)	$min(A, B)$	$A \times B$
OR ($\vee$)	$max(A, B)$	$A + B - A \times B$
NOT ($\neg$)	$1 - A$	$1 - A$

Example: Instead of "It is either raining or not," a soft logic system might say "There is a 75% chance of rain."

Exercise 5

A hospital uses a soft logic-based AI system to determine the likelihood of a patient having a serious infection. The AI assigns probability scores to different symptoms:

- Fever $(F) = 0.9$ Cough $(C) = 0.7$ Shortness of Breath (SOB) $= 0.5$

What is the probability of Fever OR Cough ($F \vee C$) using soft logic (Product-Sum)?

$$A \vee B = A + B - A \times B$$

A) 0.97
B) 0.85
C) 1.0
D) 0.89

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A) 0.97
B) 0.85
C) 1.0
D) 0.89

Answer: A — $F + C - (F \times C) = 0.9 + 0.7 - (0.9 \times 0.7) = 1.6 - 0.63 = 0.97$

Exercise 6

A hospital uses a soft logic-based AI system to determine the likelihood of a patient having a serious infection. The AI assigns probability scores to different symptoms:

- Fever $(F) = 0.9$ Cough $(C) = 0.7$ Shortness of Breath (SOB) $= 0.5$

What is the probability of (Cough AND Shortness of Breath) OR Fever: $((C \wedge \mathbf{SOB}) \vee F)$?

Soft Logic Formula:

$$A \wedge B = A \times B, \qquad A \vee B = A + B - A \times B$$

A) 0.95
B) 0.805
C) 0.935
D) 0.885

Exercise 6

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B) 0.805
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D) 0.885

Answer: C — $(C \times SOB) = 0.7 \times 0.5 = 0.35$; $(C \times SOB) \vee F = 0.35 + 0.9 - (0.35 \times 0.9) = 1.25 - 0.315 = 0.935$

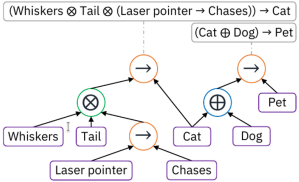
Understanding Logic Neural Networks (LNNs)

What Is a Logic Neural Network?

- A **neural-symbolic** approach for logic inference on propositional logic and first-order logic.
- Instead of relying purely on data patterns, they **enforce logical consistency** in predictions.

How does LNN perform inference?

- **Syntax tree as network structure**
- Different **activation functions** for logic connectives
- **Bi-directional message passing**



Material for Expansion:

Paper: `https://arxiv.org/pdf/2006.13155.pdf`
Tutorial:
`https://ibm.github.io/LNN/introduction.html`

Understanding Logic Neural Networks (LNNs)

How to Understand Logic Neural Networks?

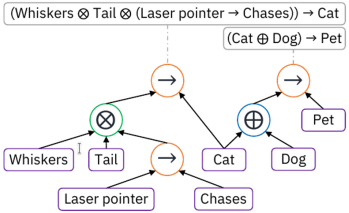
Logic formulas in the image:

Rule 1: Whiskers $\otimes$ Tail $\otimes$ (Laser pointer $\rightarrow$ Chases) $\rightarrow$ Cat

- If an entity has Whiskers **and** Tail, and if it chases a Laser pointer, then it is a **Cat**.
- $\otimes$ (multiplication operation): AND-like operation — all conditions must be satisfied.
- $\rightarrow$ (implication): Represents logical entailment — if the left-hand conditions hold, then the right-hand conclusion follows.

Rule 2: Cat $\oplus$ Dog $\rightarrow$ Pet

- If an entity is either a Cat **or** a Dog, then it is a **Pet**.
- $\oplus$ (addition operation): OR-like operation — at least one input must be true.

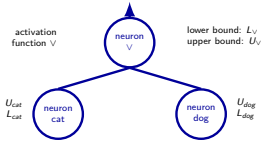


Understanding Logic Neural Networks (LNNs)

Network Structure of LNN

Key Concepts:

- **Logical Neurons in LNN:**
  - Each neuron is assigned to a node of the syntax tree, representing logical operations.
  - The network structure follows logical inference rules.
- **Probability Bounds (U, L):**
  - Each neuron maintains a **lower bound (L)** and **upper bound (U)** for the probability of its sub-formula.
  - These bounds allow handling of **uncertainty** and **partial truth values**.
- **Example:** An OR ($\vee$) operation (Each neuron in the syntax tree corresponds to a sub-formula):
  - **Neuron** $\vee$: Represents the activation function for OR operation.
  - **Neuron *cat***: Represents the probability bounds $(L_{\text{cat}}, U_{\text{cat}})$.
  - **Neuron *dog***: Represents the probability bounds $(L_{\text{dog}}, U_{\text{dog}})$.
- The final neuron outputs a new probability range $(L_{\vee}, U_{\vee})$ based on its inputs.



Understanding Logic Neural Networks (LNNs)

Workflow of Logical Neural Networks

1. Read Inputs from Data

- Axioms and Initial Bounds:
- $\neg$dog $\Rightarrow \{U_{\text{dog}} = 0,\ L_{\text{dog}} = 0\}$: If dog is false, both its upper and lower bounds are 0.
- cat $\Rightarrow \{U_{\text{cat}} = 1,\ L_{\text{cat}} = 1\}$: If cat is true, both bounds are 1.
- All other neurons initialized as $L = 0,\ U = 1$ (indicating complete uncertainty).

2. Bidirectional Message Passing Until Convergence

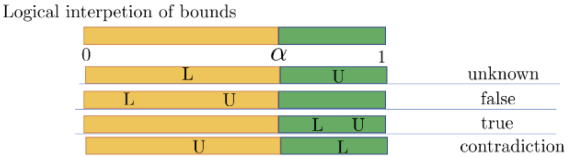
- **Upward Pass:** Updates bounds of a neuron based on its **children**.
  Example: If *dog* and *cat* influence *pet*, the truth bounds of pet are computed according to both dog and cat.
- **Downward Pass:** Updates bounds of a neuron based on its **parents**.
  Example: If *pet* is true $(U = 1, L = 1)$, then children (dog, cat) must be adjusted for consistency.

Understanding Logic Neural Networks (LNNs)

Logical Interpretation of Truth Bounds

- LNN represents truth values as bounds L and U, where $0 \leq L \leq U \leq 1$.
- Different bounds represent different logical interpretations:

Lower Bound (L)	Upper Bound (U)	Logical Meaning
0.0	1.0	Uncertain: no information available
$L \leq \alpha$	$U \leq \alpha$	False: likely false (since both bounds are below α)
$L \geq \alpha$	$U \geq \alpha$	True: likely true (since both bounds are above α)
$L > U$	—	Contradiction: inconsistent logic, conflict detected.



Understanding Logic Neural Networks (LNNs)

Three-Valued Logic

A	B	$A \wedge B$	$A \vee B$
T	T	T	T
T	U	U	T
T	F	F	T
U	U	U	U
U	F	F	U
F	F	F	F

Key Observations for AND ($\wedge$):

- If either value is False, the result is False.
- If both values are True, the result is True.
- If one value is Uncertain, the result remains Uncertain, **unless combined with False**.

Key Observations for OR ($\vee$):

- If either value is True, the result is True.
- If both values are False, the result is False.
- If one value is Uncertain, the result remains Uncertain, **unless combined with True**.

Exercise 7

If a neuron has truth bounds $L = 0.3$, $U = 0.7$, and the threshold is $\alpha = 0.5$, what does this indicate?

A) The neuron is definitely true.

B) The neuron is definitely false.

C) The neuron's truth value is uncertain.

D) The neuron has a logical contradiction.

Exercise 7

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B) The neuron is definitely false.

C) The neuron's truth value is uncertain.

D) The neuron has a logical contradiction.

Answer: C

Since $L = 0.3$ and $U = 0.7$, the truth bounds span both sides of $\alpha = 0.5$.

This means the neuron ***cannot be classified as definitely true or false and remains uncertain.***

Exercise 8

A robot assistant is trained using LNN to identify objects as "safe" or "dangerous." based on certain conditions.

The truth bounds for a given object are:

- **"Sharp" property**: $L_{\text{sharp}} = 0.2$, $U_{\text{sharp}} = 0.8$
- **"Heavy" property**: $L_{\text{heavy}} = 0.6$, $U_{\text{heavy}} = 1.0$
- **Threshold** $\alpha = 0.5$ is used for classification.
- The rule for classifying an object as dangerous: Dangerous $\leftarrow$ (Sharp $\vee$ Heavy)

Based on LNN inference, what is the truth classification of "Dangerous"?

A) The object is definitely dangerous.
B) The object is definitely safe.
C) The object is uncertain (possibly dangerous, but not guaranteed).
D) The object is in a contradictory state due to conflicting logic.

Exercise 8 – Answer: A (Definitely Dangerous)
Explanation:

- The **Heavy** property has $L = 0.6$, $U = 1.0 \rightarrow$ Both bounds $\geq \alpha = 0.5 \rightarrow$ **"Heavy" is True**.
- The **Sharp** property has $L = 0.2$, $U = 0.8 \rightarrow$ Bounds cross $\alpha \rightarrow$ **"Sharp" is Uncertain**.
- The logic rule: Dangerous $\leftarrow$ (Sharp $\vee$ Heavy). If either Sharp **OR** Heavy is true, then Dangerous must be true.
- Since **Heavy is already True**, the object is **definitely dangerous**, regardless of the uncertainty in "Sharp."

Understanding Logic Neural Networks (LNNs)

Łukasiewicz-like Logic

1. Basis Activation Function:

$$f(x) = max(0,\ min(1,\ x))$$

Ensures output remains within $[0, 1]$. Maintains differentiability for logic-based neural networks.

2. Logical AND Operation:

$$\bigwedge_{i \in I} x_i = f\left(1 - \sum_i (1 - x_i)\right)$$

Truth value is high only when **all** inputs are high.

Examples:

- $x_1 = 1,\ x_2 = 0.5$, then $\text{AND}(x_1, x_2) = f(1 - (0 + 0.5)) = f(0.5) = 0.5$
- $x_1 = 0,\ x_2 = 0$, then $\text{AND}(x_1, x_2) = f(1 - (1 + 1)) = f(-1) = 0$

Łukasiewicz-like Logic (Cont'd)

3. Logical OR Operation:

$$\bigvee_{i \in I} x_i = f\left(\sum_i x_i\right)$$

Truth value is high if **at least one** input is high.

Examples:

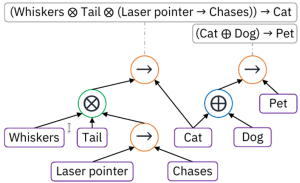
- $x_1 = 1,\ x_2 = 0.5$, then $\mathrm{OR}(x_1, x_2) = f(1 + 0.5) = f(1.5) = 1$
- $x_1 = 0,\ x_2 = 0$, then $\mathrm{OR}(x_1, x_2) = f(0 + 0) = f(0) = 0$

Understanding Logic Neural Networks (LNNs)

Inference and Convergence of LNN

LNN performs **bidirectional message passing** to update neuron truth bounds until convergence:

1. **Read inputs from data**
   - Inputs include axioms and facts, which are mapped to neuron truth bounds.
2. **Bidirectional message passing until convergence**
   - **Upward Pass:** From leaf nodes (input data) and propagates **upward**.
     E.g. Whiskers and Tail combine ($\otimes$) to influence Cat.
     Ensures that logical rules are evaluated step-by-step.
   - **Downward Pass:** From higher-level logical rules and propagates downward.
     E.g. Pet affects the truth bounds of Cat and Dog.
     Ensures logical consistency across the network.
3. **Inspect the bounds of the target neuron**
   - Once message passing converges, the truth bounds of target neurons are finalized, enabling reasoning and decision-making.



Thank you

Q&A